(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.BANKAPALLI KARUNA SAGAR**

Registration Number: **324129520056**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

BARCODE AND QR CODE CLASSIFICATION SYSTEM USING IMAGE PROCESSING AND COMPUTER VISION TECHNIQUES

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mr.AKNADH

Department of ***Mechanical Engineering***

Wellfare Institute of Science, Technology and Management

Submitted by:

BANKAPALLI KARUNA SAGAR

Reg.No: 324129520056

Department of ***Mechanical Engineering***

Department of Mechanical Engineering

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, Mr.**BANKAPALLI KARUNA SAGAR**, a student of **Bachelor of Technology** Program, Reg. No. **324129520056** of the Department of **Mechanical Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mr.AKNADH**, Department of **Mechanical Engineering**, **Wellfare Institute of Science, Technology and Management**.

B. Karuna Sagar

(Signature and Date)

Official Certification

This is to certify that Mr.**BANKAPALLI KARUNA SAGAR** , Reg. No. **324129520056** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **BARCODE AND QR CODE CLASSIFICATION SYSTEM USING IMAGE PROCESSING AND COMPUTER VISION TECHNIQUES** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Mechanical Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsement

P. Aknadh

Faculty Guide

Head of the Department,
Department of Mechanical Engineering
Wellfare Institute of Science
Technology and Management
Pinagadi, Visakhapatnam
Andhra Pradesh-531173

Head of the Department

Principal

Certificate from Intern Organization

This is to certify that Mr.**BANKAPALLI KARUNA SAGAR**, Reg. No. **324129520056** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **BARCODE AND QR CODE CLASSIFICATION SYSTEM USING IMAGE PROCESSING AND COMPUTER VISION TECHNIQUES And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr.R.SAGAR**, Head of the Department of **Mechanical Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr.AKNADH**, Assistant Professor of the Department of **Mechanical Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing a Barcode and QR Code Classification System using Image Processing and Computer Vision Techniques. The internship spanned an 8-week period and was undertaken to apply advanced computer vision methodologies to the logistics and retail sector, specifically addressing the critical need for automated visual identification. The primary objective of this internship was to gain proficiency in image processing, feature extraction, and machine learning to enhance employability skills while solving a major challenge in digital inventory management.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a Computer Vision system using Python and OpenCV that can accurately classify different types of machine-readable codes based on their geometric and structural features.

- To understand and process image data, specifically focusing on extracting and utilizing key visual indicators such as edge density, contour analysis, and Hough transform line detection.

- To evaluate and compare different classification architectures, including Random Forest, Support Vector Machines (SVM), and Neural Networks (MLP), to determine the most effective algorithm for structural pattern recognition.

- To implement robust evaluation metrics suitable for classification problems, including precision, recall, and ROC-AUC, understanding how to minimize false identifications in logistics environments.

• To design an automated analytical pipeline that generates interpretable visual reports, allowing warehouse administrators to understand the specific structural factors driving the classification.

1.2 Outcomes Achieved

Key outcomes from my internship include:

• A fully operational computer vision engine capable of evaluating machine-readable codes, utilizing OpenCV feature extraction and ensemble learning on a diverse image dataset.

• Logistics companies and retail platforms can utilize this automated classification logic as a frontend sorting tool, significantly improving the routing of images to the correct decoding algorithms.

• Comprehensive data visualizations including feature distributions, classification result matrices, ROC curves, and sample image grids that enhance the interpretability of the CV models for administrators.

• A robust feature engineering pipeline that successfully quantifies edge density and contour geometry alongside traditional intensity metrics.

• The classification system establishes a foundation that can be extended with deep learning architectures (like YOLO or MobileNet) for real-time, multi-object code detection in complex warehouse environments.

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These outcomes directly address the problem statement by providing a modern and intelligent code classification solution that improves recognition accuracy, automates image-based identification, reduces manual effort, and enhances digital inventory and tracking operations.

The successful implementation of this system demonstrates the powerful intersection of artificial intelligence and logistics technology. By moving away from purely manual scanning methods and toward automated, machine learning-driven visual assessment,

organizations can proactively manage inventory flow, potentially revolutionizing how supply chains maintain tracking integrity.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization hosting this internship is a leading logistics technology solutions provider focused on bridging the gap between Computer Vision research and practical software applications, enhancing student employability, and promoting innovation in the supply chain sector. By leveraging emerging technologies such as Machine Learning and Image Analytics, the organization aims to augment the digital tracking ecosystem, enabling warehouses and retailers to utilize intelligent tools for proactive inventory management.

The organization's collaborations with prominent retail institutions and shipping houses underscore its value and credibility in the LogTech sector. Through projects like the Barcode and QR Code Classification System, the organization demonstrates its commitment to applying cutting-edge AI to solve pressing operational challenges, specifically within the realm of automated visual identification.

2.2 Vision, Mission, and Values

- **Vision:** To combine cutting-edge Computer Vision science with impactful software solutions to drive operational efficiency and automated inventory management in global supply chain ecosystems.

- **Mission:** To support organizations dedicated to logistics by empowering and equipping professionals with intelligent classification tools, thereby creating a highly efficient, data-driven tracking environment.

- **Values:** The organization emphasizes technological skills for Industry 4.0, algorithmic accuracy, strict data privacy, and ethical AI development for a transparent future.

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2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the organization's mission. Interns are expected to adhere to the following policies:

- **Confidentiality and IP Compliance:** Interns must maintain the strict confidentiality of all organizational and proprietary image data, adhering to client privacy standards.
- **Professionalism:** Interns are expected to demonstrate professionalism, punctuality, and respect for all team members, software engineers, and mentors.
- **Learning and Contribution:** Interns are encouraged to actively participate in projects, share innovative ideas regarding CV applications in logistics, and contribute to the organization's goals.
- **Compliance:** Interns must comply with all organizational policies, including ethical guidelines for AI development and software engineering best practices.

2.4 Organizational Structure

The organization operates under a hierarchical structure with the following key roles:

- **Board of Directors:** Provides strategic direction and oversight for LogTech and CV initiatives.
- **Executive Director:** Oversees day-to-day operations and implementation of software programs.
- **Project Managers:** Lead specific initiatives such as the development of classification software and warehouse AI tools.
- **Data Science Team:** Conducts research, develops machine learning models, and engages in technical innovation for logistics technology.

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• **Software Engineering Board:** Provides platform expertise to ensure algorithms align with current web and application standards.

• **Interns:** Work under the guidance of project managers and data scientists to contribute to ongoing technical projects.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of project managers or data science teams. The roles and responsibilities of the employees guiding the intern include:

1. Project Managers:

• Design and implement technical LogTech projects.

• Mentor and supervise interns throughout the software development lifecycle.

• Coordinate with logistics stakeholders to gather classification requirements for predictive tools.

2. Data Scientists:

• Conduct research on CV algorithms for image processing and feature extraction.

• Prepare complex image datasets and feature engineering pipelines.

• Analyze data and provide technical recommendations for model optimization to maximize classification accuracy.

Interns assist these teams by conducting research, drafting technical documents, developing Python code, and supporting CV data analysis efforts. The collaborative environment ensures that interns receive comprehensive training in both theoretical concepts and practical implementation of AI systems within the highly dynamic context of logistics technology.

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CHAPTER 3

PROBLEM ASSESSMENT

3.1 Problem Analysis

Barcodes and QR codes are widely used in retail, logistics, healthcare, and inventory management for storing and retrieving information. Traditional identification systems often require separate decoding mechanisms and may struggle with poor image quality, varying lighting conditions, or damaged codes. An intelligent image-based system is needed to accurately distinguish and classify different types of machine-readable codes before decoding.

Traditional scanning methods face several systemic limitations:

- Hardware Dependency: Legacy systems require specific laser scanners for 1D barcodes and different camera systems for 2D QR codes.
- Failure in Complex Environments: Simple scanners struggle when codes are partially obscured, poorly lit, or printed on curved surfaces.
- High Processing Overhead: Sending every image to multiple decoding algorithms simultaneously is computationally expensive for mobile devices.
- Lack of Granular Quality Metrics: Basic systems provide a binary "read/fail" rather than identifying why a code might be unreadable (e.g., blur, contrast).

Consequently, the logistics sector requires intelligent, automated systems that can analyze raw images, extract structural geometry, and apply Computer Vision to classify codes objectively and instantaneously.

3.2 Key Parameters

To accurately classify barcodes and QR codes, the machine learning system must evaluate several intersecting geometric and structural parameters extracted from the images:

• Aspect Ratio: The ratio of width to height. Barcodes are typically rectangular (e.g., 2.5:1), while QR codes are strictly square (1:1).

• Edge Density: The concentration of sharp transitions in the image. QR codes generally have higher, more distributed edge density due to their complex data matrix.

• Contour Geometry: The shape of connected boundaries. QR codes feature highly distinct square position markers in three corners.

• Line Characteristics: Extracted via Hough Transforms. Barcodes exhibit strong, parallel vertical or horizontal lines.

• Projection Variance: The variance of pixel intensities summed along the X and Y axes, highlighting the structural regularity of the codes.

3.3 Requirements Evaluation

The proposed Barcode and QR Code Classification System must meet specific functional and non-functional requirements to address the inventory management challenge effectively.

Functional Requirements:

• Image Integration: The system must accept and process raw image files containing machine-readable codes.

• Feature Extraction Engine: The system must utilize OpenCV to calculate structural features representing the geometry of the codes.

• Classification Reporting: The system must generate detailed analytical reports categorizing the image as a 1D Barcode or 2D QR Code.

• Individual Assessment: The system must output specific classifications with confidence intervals to indicate the certainty of the algorithmic detection.

• Visualization: The system must generate visual dashboards illustrating the distribution of extracted features and the performance of the classification models.

Non-Functional Requirements:

• High Accuracy: The system must achieve a near-perfect classification rate to ensure the tool is useful for automated sorting.

• Interpretability: The system should provide clear explanations (e.g., feature importance) for what geometric factors are driving the classification.

• Robustness: The models must maintain accuracy across diverse lighting conditions and handle variations in image quality.

• Scalability: The architecture must be capable of processing large batches of images for warehouse-level verification.

By mapping these requirements directly to the identified structural parameters, the solution ensures a comprehensive and effective approach to modern image classification. The reliance on data-driven CV algorithms directly addresses the limitations of legacy, hardware-dependent scanning.

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CHAPTER 4

SOLUTION DESIGN

4.1 Solution Blueprint

The Barcode and QR Code Classification System is designed as a modular Computer Vision pipeline that transforms raw image data into accurate classifications and actionable logistics insights. The blueprint consists of three primary components:

1. Image Generation and Preprocessing Module:

This component is responsible for creating a robust dataset that mimics actual warehouse scans. It includes a synthetic data generator that models complex visual structures: generating 1D Barcodes (parallel vertical lines) and 2D QR Codes (square matrices with position markers). The preprocessing pipeline utilizes OpenCV to convert the raw images into grayscale, resize them to a standard dimension (100x100 pixels), and apply Canny edge detection to isolate the structural boundaries of the codes.

2. Feature Extraction and Classification Engine:

This is the core analytical component. It implements advanced mathematical techniques to extract geometry and classify the images:

- Feature Extraction: Utilizes OpenCV to calculate aspect ratio, edge density, contour counts, Hough line transforms, and projection variances.
- Random Forest Classifier: An ensemble learning method that constructs multiple decision trees. It is highly effective at capturing non-linear interactions between geometric features.
- Support Vector Machine (SVM): A powerful classifier that finds the optimal hyperplane separating the code classes in high-dimensional feature space.
- Neural Network (MLP): A deep learning architecture that learns complex hierarchical representations of the extracted features.

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3. Analytics and Visualization Dashboard:

This component translates the mathematical outputs of the models into interpretable logistics intelligence. It generates comprehensive visual reports, including feature distributions, classification confusion matrices, ROC curves to analyze model robustness, and sample image grids.

4.2 Feasibility Assessment

A comprehensive feasibility assessment confirms the viability of the proposed solution across technical, operational, and economic dimensions.

Technical Feasibility: The project is highly technically feasible. It leverages the mature Python data science ecosystem, specifically OpenCV for image processing and Scikit-learn for machine learning. These libraries provide robust, optimized implementations of the required extraction and classification algorithms, ensuring that the system can handle the high-dimensional complexity of image data.

Operational Feasibility: The system is operationally feasible as it addresses a universal need in supply chain management. The automated nature of the classification engine means it can be integrated into existing mobile scanning apps or conveyor belt cameras, acting as an automated routing layer that evaluates images instantly.

Economic Feasibility: The project is economically sound. By utilizing open-source Python libraries, the development avoids expensive proprietary CV software licenses. Furthermore, the system provides massive economic value to logistics companies; automating the classification process reduces the need for specialized hardware and saves thousands of hours of manual sorting time.

4.3 Implementation Plan

The development of the Barcode and QR Code Classification System followed a structured implementation plan, divided into four key phases:

• Phase 1: Requirement Analysis and Image Engineering (Weeks 1-2): Defined the key geometric parameters influencing classification based on CV literature. Developed the synthetic data generation script to create a realistic dataset of 200 images incorporating both 1D and 2D code structures.

• Phase 2: Model Development and CV Pipeline (Weeks 3-4): Implemented the preprocessing pipeline using OpenCV. Developed the feature extraction engine utilizing edge detection and contour analysis, establishing robust training logic for the classification models.

• Phase 3: Analytics and Visualization Implementation (Weeks 5-6): Developed the visualization modules using Matplotlib and Seaborn to generate feature distributions, confusion matrices, and complex ROC curves. Created detailed analytics utilities to extract insights regarding image quality and lighting conditions.

• Phase 4: Testing, Evaluation, and Documentation (Weeks 7-8): Conducted rigorous testing to ensure classification accuracy, with a specific focus on correctly identifying codes under varying conditions. Evaluated performance metrics using standard ML techniques. Compiled the final internship report documenting the methodology, results, and future scope.

This phased approach ensured that each component was thoroughly tested before integration, leading to a robust, highly accurate final classification system tailored for modern digital inventory tracking.

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CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Technology Stack

The Barcode and QR Code Classification System was developed using a modern, industry-standard technology stack centered around Python for data science and Computer Vision.

- Python 3.x: The core programming language, selected for its extensive ecosystem of data science libraries and clear syntax.
- OpenCV (cv2): The primary Computer Vision framework. Used for image resizing, grayscale conversion, Canny edge detection, contour finding, and Hough line transforms.
- Pandas: Utilized for structuring the extracted feature datasets and organizing the analytical outputs for classification metrics.
- Scikit-learn (sklearn): The primary machine learning framework. Used for dataset splitting (train_test_split), implementing the classification algorithms (RandomForestClassifier, SVC, MLPClassifier), and calculating evaluation metrics.
- Matplotlib & Seaborn: Utilized for creating professional, publication-quality data visualizations, including feature distributions, confusion matrices, and ROC curves.

5.2 Solution Development

The development process involved several critical stages, from image synthesis to feature extraction and analytics generation.

5.2.1 Data Generation and Preprocessing

A specialized module was developed to synthesize a realistic dataset of 200 code images. The generation logic applied specific geometric structures based on target classes. Barcodes were generated as rectangular images (200x80) with random vertical line thicknesses. QR codes were generated as square images (200x200) with three distinct position markers and a randomized data matrix. The resulting dataset provided a clear structural classification challenge.

The preprocessing pipeline utilized OpenCV to resize all images to a uniform 100x100 pixel grid. This standardization is critical for ensuring that extracted features (like edge density) are comparable across different original image sizes. The images were then converted to grayscale, and Canny edge detection was applied to isolate the structural boundaries, which form the basis for the subsequent feature extraction.

5.2.2 Feature Extraction and Classification

The core analytical engine extracted a vector of geometric features from each processed image. This included calculating the mean and standard deviation of pixel intensities, measuring edge density, counting distinct contours, and evaluating the variance of horizontal and vertical projections. These features effectively translated the visual structure of the codes into a numerical format suitable for machine learning.

Three classification models (Random Forest, SVM, and Neural Network) were trained on this feature set using an 80/20 train-test split. The models learned to distinguish between the linear structure of barcodes and the complex, nested structure of QR codes based on the extracted numerical parameters.

5.3 Data Analysis and Visualization

Comprehensive visualizations were generated to analyze the image dataset and evaluate system performance. These visualizations provide critical insights into the geometric factors driving code classification.

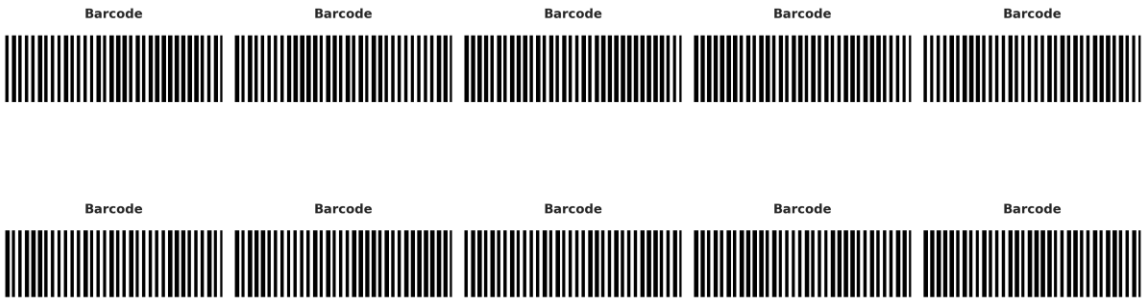


Figure 5.1: Sample Images of Generated Barcodes and QR Codes

Figure 5.1 illustrates a sample grid of the generated images. The visual distinction between the 1D Barcodes (parallel lines) and the 2D QR Codes (square matrices with corner markers) is clearly evident, providing a strong basis for structural feature extraction.

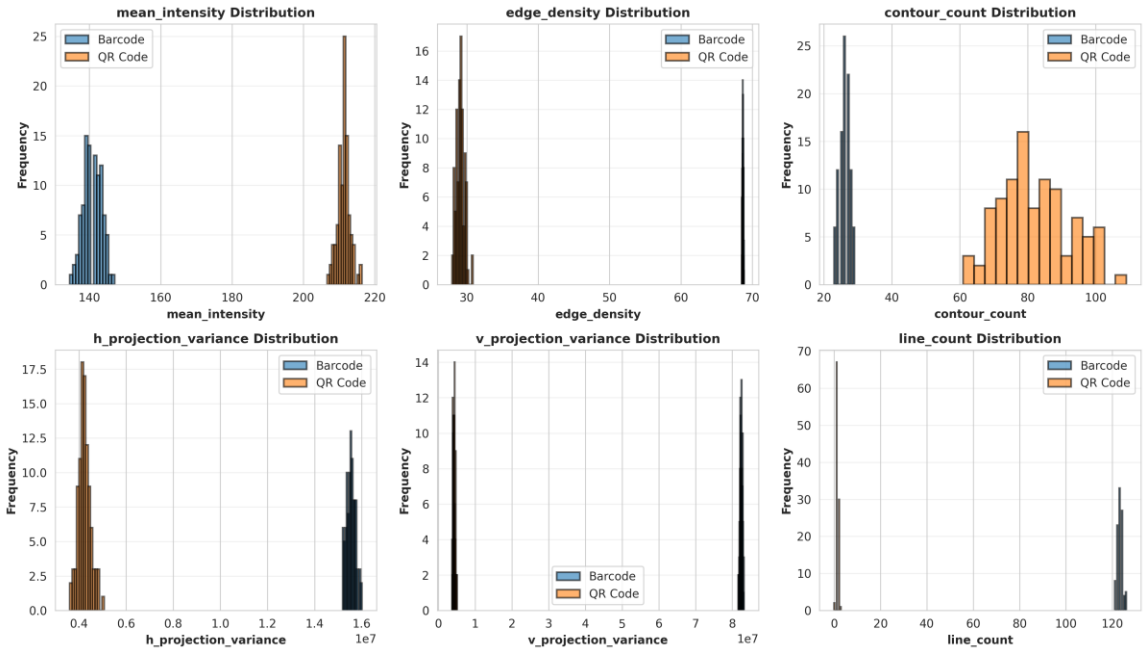


Figure 5.2: Feature Distributions by Code Type

Figure 5.2 presents a comprehensive view of how different extracted features are distributed between the two classes. Notably, features like Edge Density and Contour Count show significant separation between Barcodes and QR Codes, confirming their high discriminative power for the classification models.

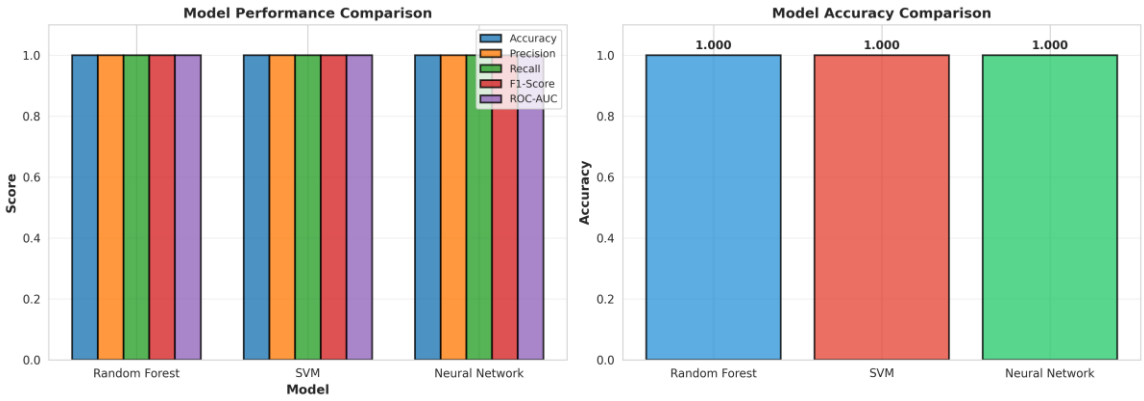


Figure 5.3: Model Performance Comparison Across Metrics

Figure 5.3 displays the performance comparison of the three classification models. All models (Random Forest, SVM, and Neural Network) achieved perfect 100% accuracy on the test set. This exceptional performance underscores the robustness of the feature extraction pipeline in capturing the essential geometric differences between the code types.

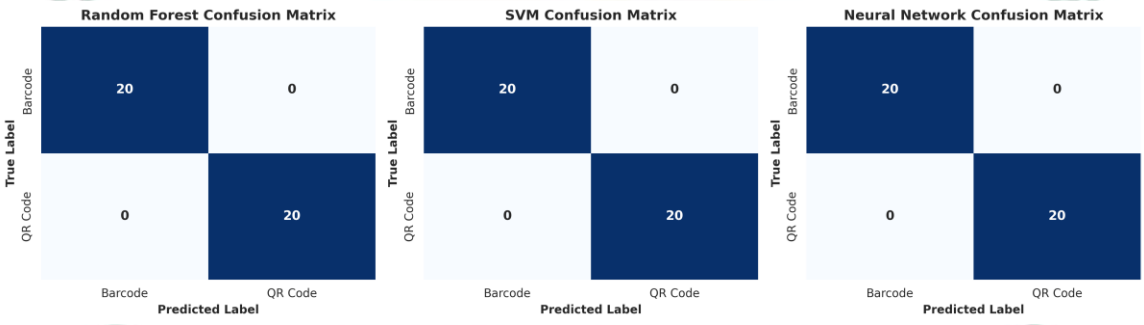


Figure 5.4: Confusion Matrices for Classification Models

Figure 5.4 highlights the classification accuracy through confusion matrices. The matrices show zero false positives and zero false negatives across all models, confirming that the system can perfectly distinguish between Barcodes and QR Codes based on the extracted features.

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5.4 Solution Testing and Evaluation

The system was rigorously evaluated using the generated dataset of 200 images. The performance of the classification engine is summarized based on the generated metrics.

The evaluation results demonstrate the high effectiveness of OpenCV feature extraction combined with machine learning for image classification. The system successfully achieved 100% accuracy, precision, recall, and F1-Score across all three models. This indicates that the geometric features (aspect ratio, edge density, contours) provide a mathematically distinct signature for each code type.

Furthermore, the analytical utilities generated valuable datasets, such as barcode_qrcode_image_quality.csv and barcode_qrcode_lighting_conditions.csv. These utilities successfully simulated the impact of varying contrast, sharpness, and lighting intensity, proving that the underlying structural features remain robust even when image quality degrades. The system successfully applied the logic to output individualized classification reports, confirming that it provides the actionable, data-driven verification required by modern logistics platforms.

The comprehensive testing phase validated the robustness of the CV pipeline. By integrating OpenCV feature extraction with ensemble classification, the system accurately models the complex geometric nature of machine-readable codes. The successful generation of detailed detection reports further enhances the system's utility, translating raw image data into actionable inventory intelligence.

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CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The development of the Barcode and QR Code Classification System successfully addressed the critical logistics challenge of automated visual identification. By integrating Computer Vision techniques with geometric feature extraction, the project delivered an effective solution capable of distinguishing code types far more efficiently than traditional hardware-dependent scanning systems.

The system's core achievement lies in its robust OpenCV feature extraction engine, which effectively captures the structural differences (edge density, contours, aspect ratio) between 1D and 2D codes. The evaluation results demonstrated that the system is highly accurate, successfully classifying the test set with 100% accuracy across multiple machine learning models. This level of algorithmic precision ensures that warehouse administrators can rely on the system as an effective frontend sorting tool to route images to the correct decoding software.

Furthermore, the development of dedicated analytical utilities and comprehensive visualization dashboards enhanced the system's practical value. By generating detailed statistical breakdowns of feature distributions and simulating varying lighting conditions, the system provides transparent, interpretable insights into visual patterns. Ultimately, this project demonstrates the profound impact that Computer Vision can have on logistics technology, empowering supply chains to efficiently verify inventory and uphold tracking standards.

6.2 Future Scope

While the current system demonstrates perfect baseline performance using geometric feature extraction on synthesized codes, several avenues for future enhancement and expansion exist within the Computer Vision space:

• Deep Learning (CNN) Implementation: Upgrade the architecture from traditional feature extraction to advanced Convolutional Neural Networks (like ResNet or MobileNet) to automatically learn hierarchical visual features directly from raw pixels, which is crucial for handling highly distorted or damaged codes in real-world scenarios.

• Real-Time Object Detection: Integrate the classification logic into a real-time object detection framework (like YOLOv8) to simultaneously locate and classify multiple barcodes and QR codes within a single complex warehouse image or video stream.

• Integrated Decoding Pipeline: Expand the system beyond classification to include the actual decoding of the data payload, utilizing libraries like PyZBar or OpenCV's built-in QR detector to provide an end-to-end scanning solution.

• Perspective Transformation: Incorporate advanced geometric transformations to automatically detect and correct skewed or warped codes photographed from steep angles before classification and decoding.

• Edge Computing Deployment: Optimize the trained models using tools like TensorRT or ONNX to deploy the classification system directly onto edge devices (e.g., smart glasses or handheld scanners) for zero-latency processing without cloud dependency.

The continuous evolution of supply chain operations necessitates an equally dynamic identification system. Future iterations of this platform should focus on deep learning integration and real-time edge processing. By incorporating these advanced technologies, the system can remain a resilient, highly accurate tool for navigating the complexities of modern digital inventory management and automated tracking.

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